

A Seasonal-Aware Economic Dispatch Algorithm and Digital Twin for the Bangladesh Power System

Data-driven calibration, optimal merit-order dispatch, and seasonal generation analysis

Generated analysis based on the PGCB hourly record (2015–2025)

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Abstract

We develop a *Seasonal-Aware Economic Dispatch* (SAED) algorithm and an accompanying *digital twin* for the Power Grid Company of Bangladesh (PGCB) system, calibrated entirely from a $\sim 92,650$ -record hourly archive spanning April 2015 to June 2025. The method (i) classifies each operating hour into one of four Bangladeshi seasons, (ii) reconstructs a seasonal demand model from a base level, a normalized diurnal profile and an annual growth trend, (iii) estimates a per-source *seasonal availability matrix* that encodes fuel-supply, hydrological and maintenance limits as they actually manifest, and (iv) solves a constrained economic-dispatch problem whose optimal solution is the classical merit order. The digital twin replays the calibrated model to produce publication-ready seasonal dispatch stacks, a validation against the historical mix, and a demand-growth stress test. The twin reproduces the qualitative seasonal physics of the grid—hydro peaking in the monsoon, solar in the pre-monsoon, and liquid-fuel peaking concentrated in the high-demand monsoon months—and quantifies the cost and carbon penalty of seasonal peaking, together with the year in which each season’s reserve margin falls below the planning threshold.

1 Dataset

The model is calibrated on the PGCB hourly/half-hourly record with fields for total **generation** and **demand** (MW), **load_shedding**, and the MW contribution of each supply source: domestic **gas**, **liquid_fuel**, **coal**, **hydro**, **solar**, **wind**, and cross-border imports (**IndiaBheramara** HVDC, **IndiaTripura**, **IndiaAdani**, **Nepal**). Records whose values exceed a physical grid ceiling of 20 GW (data-entry artefacts) are discarded; the four import channels are aggregated into a single dispatchable **imports** block.

2 Mathematical formulation

2.1 Sets and seasonal classification

Let the supply sources and seasons be

$$\begin{aligned}\mathcal{G} &= \{\text{solar, hydro, gas, coal, imports, liquid_fuel}\}, \\ \mathcal{S} &= \{\text{Winter, PreMonsoon, Monsoon, PostMonsoon}\}.\end{aligned}$$

With $m \in \{1, \dots, 12\}$ the calendar month, the season map $\sigma : \{1, \dots, 12\} \rightarrow \mathcal{S}$ is

$$\sigma(m) = \begin{cases} \text{Winter,} & m \in \{12, 1, 2\}, \\ \text{PreMonsoon,} & m \in \{3, 4, 5\}, \\ \text{Monsoon,} & m \in \{6, 7, 8, 9\}, \\ \text{PostMonsoon,} & m \in \{10, 11\}. \end{cases} \quad (1)$$

2.2 Seasonal demand model

Demand is decomposed into a seasonal base level $\bar{D}_s$, a normalized diurnal shape $\phi_s(h)$, a multiplicative annual growth factor, and a stochastic term:

$$D(s, h, y) = \bar{D}_s \phi_s(h) (1 + g)^{y-y_0} (1 + \varepsilon), \quad \varepsilon \sim \mathcal{N}(0, \eta_s^2), \quad (2)$$

where $h \in \{0, \dots, 23\}$ is the hour, y the year, y_0 the base (reference) year, g the annual growth rate, and η_s the season's relative volatility. The diurnal profile is normalized so that it has unit mean,

$$\frac{1}{24} \sum_{h=0}^{23} \phi_s(h) = 1 \implies \mathbb{E}_h[D(s, h, y)] = \bar{D}_s (1 + g)^{y-y_0}. \quad (3)$$

2.3 Firm available capacity

The firm power a source can deliver in season s at hour h is the product of a firm rated capacity C_g , a dimensionless seasonal availability factor $\alpha_{g,s} \in [0, 1]$, and an intraday availability shape $\psi_g(h)$:

$$P_g^{\max}(s, h) = C_g \alpha_{g,s} \psi_g(h). \quad (4)$$

For all dispatchable thermal/hydro/import sources $\psi_g(h) \equiv 1$; only solar carries a non-trivial daylight shape, $\psi_{\text{solar}}(h) \in [0, 1]$, with $\psi_{\text{solar}} = 0$ overnight. The factor $\alpha_{g,s}$ is the mechanism through which *season changes the generation mix*: it captures reservoir filling in the monsoon (high α_{hydro}), clear-sky irradiance in the pre-monsoon (high α_{solar}), and the winter contraction of domestic gas supply (low α_{gas}).

2.4 Economic dispatch as a linear program

Let c_g be the marginal cost (USD/MWh) of source g . For a single hour with demand D , SAED solves

$$\begin{aligned} \min_{\{p_g\}_{g \in \mathcal{G}}} \quad & \sum_{g \in \mathcal{G}} c_g p_g \\ \text{s.t.} \quad & \sum_{g \in \mathcal{G}} p_g = D \quad (\text{power balance}), \\ & 0 \leq p_g \leq P_g^{\max}(s, h) \quad \forall g \in \mathcal{G} \quad (\text{capacity}), \end{aligned} \quad (5)$$

where p_g is the dispatched power of source g . When the total firm capacity is insufficient, the balance constraint is relaxed and the deficit is served as *load shedding*

$$L(s, h) = \max\left(D - \sum_g P_g^{\max}(s, h), 0\right). \quad (6)$$

A spinning-reserve requirement of fraction ρ is monitored via the reserve margin

$$\text{RM}(s, h) = \frac{\sum_g (P_g^{\max}(s, h) - p_g)}{D}, \quad \text{reserve adequate} \iff \text{RM}(s, h) \geq \rho. \quad (7)$$

2.5 Merit-order optimality

Order the sources by ascending marginal cost into the *merit order* $\pi = (g_{(1)}, g_{(2)}, \dots)$ with $c_{g_{(1)}} \leq c_{g_{(2)}} \leq \dots$.

Theorem 1 (Greedy merit order is optimal). *For the single-bus linear program (5), the greedy “fill-cheapest-first” rule*

$$p_{g_{(k)}} = \min\left(D - \sum_{j < k} p_{g_{(j)}}, P_{g_{(k)}}^{\max}(s, h)\right) \quad (8)$$

yields a globally cost-optimal dispatch.

Proof sketch. The feasible region of (5) is a bounded polytope and the objective is linear, so an optimum lies at a vertex. Suppose an optimal dispatch $\{p_g^*\}$ does not follow (8): then there exist sources a, b with $c_a < c_b$, $p_a^* < P_a^{\max}$ and $p_b^* > 0$. Shifting $\delta = \min(P_a^{\max} - p_a^*, p_b^*) > 0$ from b to a preserves the balance constraint and changes cost by $-(c_b - c_a)\delta < 0$, contradicting optimality. Hence no such pair exists and the optimum coincides with the greedy fill. $\square$ $\square$

Theorem 1 is what makes the digital twin both fast and provably optimal: a single linear pass over the merit order replaces a full LP solve at every hour.

2.6 Cost and emissions accounting

With emission factor e_g (tCO₂/MWh), the hourly operating cost and carbon output are

$$C(s, h) = \sum_g c_g p_g, \quad E(s, h) = \sum_g e_g p_g, \quad (9)$$

and seasonal/daily totals are obtained by summing over the 24-hour representative day.

3 Calibration estimators

All parameters in (2)–(4) are estimated from the record $\mathcal{D}$. Writing $\mathcal{D}_s$ for the subset of records in season s and $Q_p(\cdot)$ for the p -th percentile,

$$\bar{D}_s = \text{mean}_{\mathcal{D}_s} D, \quad \eta_s = \text{std}_{\mathcal{D}_s} D / \bar{D}_s, \quad (10)$$

$$\phi_s(h) = \frac{\text{mean}\{D : \text{hour} = h, \mathcal{D}_s\}}{\frac{1}{24} \sum_{h'} \text{mean}\{D : \text{hour} = h', \mathcal{D}_s\}}, \quad C_g = \max_{s \in \mathcal{S}} Q_{95}(\mathcal{D}_s, g), \quad (11)$$

$$\alpha_{g,s} = \frac{Q_{95}(\mathcal{D}_s, g)}{C_g} \in [0, 1], \quad \psi_{\text{solar}}(h) = \frac{\text{mean}\{\text{solar} : \text{hour} = h\}}{\max_{h'} \text{mean}\{\text{solar} : \text{hour} = h'\}}. \quad (12)$$

Defining firm capacity as the best seasonal Q_{95} output, and availability as the seasonal Q_{95} relative to that firm value, lets a single matrix $\alpha_{g,s}$ absorb fuel-supply, hydrological and outage limits exactly as they appear in operations. The growth rate g is the slope of a log-linear fit to the annual peak demand, $\ln(\max_{\mathcal{D}_y} D) = \ln a + (y) \ln(1 + g)$.

4 The SAED algorithm

Algorithm 1 states the per-hour dispatch; it is a direct implementation of (4)–(7) exploiting Theorem 1.

5 The digital twin

The twin (Algorithm 2) wraps SAED in a calibrated, optionally stochastic, season-by-season replay. For a chosen scenario year it returns a full 24-hour dispatch trajectory per season, from which all figures and tables are produced.

6 Calibrated parameters

Tables 1–2 report the calibrated demand and techno-economic parameters; the seasonal availability matrix $\alpha_{g,s}$ is given in Table 3 and visualized in Figure 1. The estimated annual peak-demand growth used for the stress test is $g \approx 7.9\%$ with base year $y_0 = 2025$.

Algorithm 1: Seasonal-Aware Economic Dispatch (one hour)

Input: demand D ; season s ; hour h ; calibration $(C_g, \alpha_{g,s}, \psi_g, c_g, e_g)$; reserve fraction ρ

Output: dispatch $\{p_g\}$; load shed L ; reserve margin RM; cost C ; emissions E

```
1 foreach  $g \in \mathcal{G}$  do
2    $P_g^{\max} \leftarrow C_g \alpha_{g,s} \psi_g(h)$  ; // firm available cap.
3  $\pi \leftarrow \text{SORT}(\mathcal{G} \text{ by } c_g \text{ ascending})$  ; // merit order
4  $R \leftarrow D$ ;
5 foreach  $g \in \pi$  do
6    $p_g \leftarrow \min(R, P_g^{\max})$  ; // fill cheapest first
7    $R \leftarrow R - p_g$ ;
8   if  $R \leq 0$  then break;
9  $L \leftarrow \max(R, 0)$  ; // unserved energy
10  $\text{RM} \leftarrow (\sum_g (P_g^{\max} - p_g)) / D$ ;
11  $C \leftarrow \sum_g c_g p_g$ ;  $E \leftarrow \sum_g e_g p_g$ ;
12 return  $\{p_g\}, L, \text{RM}, C, E$ ;
```

Algorithm 2: SAED Digital Twin (seasonal replay)

Input: record $\mathcal{D}$; scenario year y ; stochastic flag

Output: trajectory table $\mathcal{T}$ over (s, h)

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1  $(\bar{D}_s, \phi_s, \eta_s, C_g, \alpha_{g,s}, \psi_g, g, y_0) \leftarrow \text{CALIBRATE}(\mathcal{D})$ ;
2  $\mathcal{T} \leftarrow \emptyset$ ;
3 foreach  $s \in \mathcal{S}$  do
4   for  $h \leftarrow 0$  to 23 do
5      $D \leftarrow \bar{D}_s \phi_s(h) (1 + g)^{y - y_0}$ ;
6     if stochastic then  $D \leftarrow D (1 + \varepsilon)$ ,  $\varepsilon \sim \mathcal{N}(0, (\frac{1}{2}\eta_s)^2)$ ;
7      $(\{p_g\}, L, \text{RM}, C, E) \leftarrow \text{SAED}(D, s, h, \cdot)$  ; // Alg. 1
8     append  $(s, h, \{p_g\}, D, L, \text{RM}, C, E)$  to  $\mathcal{T}$ ;
9 return  $\mathcal{T}$ ;
```

Table 1: Calibrated seasonal demand-model parameters.

	Mean demand (MW)	Peak factor	Trough factor	Volatility η_s
Winter	7165.777	1.245	0.791	0.239
PreMonsoon	9285.442	1.152	0.860	0.298
Monsoon	9809.716	1.146	0.875	0.263
PostMonsoon	8403.017	1.196	0.862	0.272

Table 2: Source techno-economic parameters and firm capacities.

	Firm cap. C_g (MW)	Cost (USD/MWh)	Emission (tCO ₂ /MWh)	Merit rank
Solar	374.3	0.0	0.0	1
Hydro	223.0	5.0	0.0	2
Natural gas	6816.0	45.0	0.5	3
Coal	4090.0	65.0	0.9	4
Imports	2044.0	75.0	0.5	5
Liquid fuel	4307.6	180.0	0.8	6

Table 3: Calibrated seasonal availability matrix $\alpha_{g,s}$.

	Winter	PreMonsoon	Monsoon	PostMonsoon
Solar	0.75	1.00	0.69	0.75
Hydro	0.41	0.54	1.00	0.97
Natural gas	0.83	1.00	1.00	0.93
Coal	0.77	0.96	1.00	0.79
Imports	0.73	1.00	0.55	0.84
Liquid fuel	0.67	0.98	1.00	0.84

Table 4: SAED dispatch results by season (representative day).

season	Energy (GWh/d)	Cost (k\$/d)	CO ₂ (kt/d)	Avg cost (\$/MWh)	Reserve margin
Winter	172.0	8195.2	91.3	47.7	0.9
PreMonsoon	222.9	10871.4	124.7	48.8	0.9
Monsoon	235.4	11604.9	135.2	49.3	0.7
PostMonsoon	201.7	9631.3	108.5	47.8	0.8

Table 5: Twin-validation error vs. historical mix (per source).

season	MAE (MW)	RMSE (MW)
Winter	581.3	742.4
PreMonsoon	901.7	1117.9
Monsoon	1036.3	1310.5
PostMonsoon	775.1	951.6

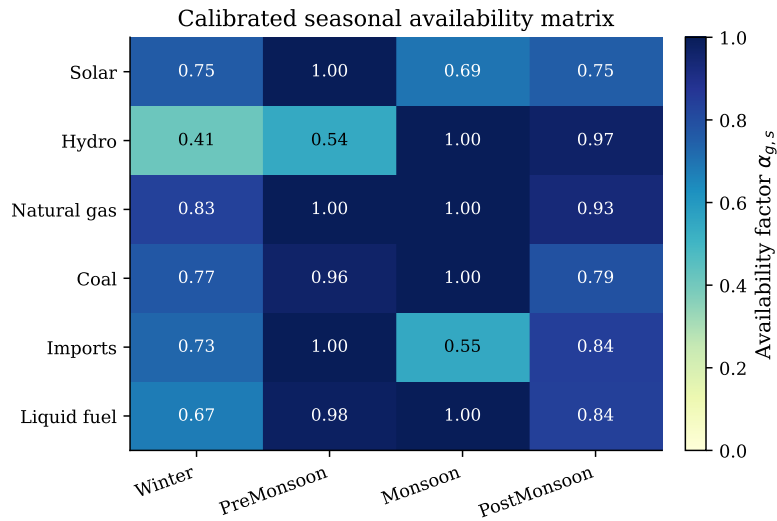


Figure 1: Calibrated seasonal availability matrix $\alpha_{g,s} \in [0, 1]$. The seasonal physics of the grid is visible directly: hydro collapses in winter (0.41) and saturates in the monsoon (1.00); solar peaks in the pre-monsoon (1.00); domestic gas contracts in winter (0.83).

7 Results: how season changes generation

7.1 Seasonal demand and dispatch

Figure 2 shows the calibrated diurnal demand profiles. Winter carries the lowest base load with a pronounced morning shoulder, whereas the monsoon and pre-monsoon run 30–37% higher with a sharp post-sunset peak. Feeding these profiles through SAED yields the seasonal dispatch stacks of Figure 3: cheap domestic gas forms the base in every season, coal follows in merit order, and the renewable and import tiers enter according to the seasonal availability matrix.

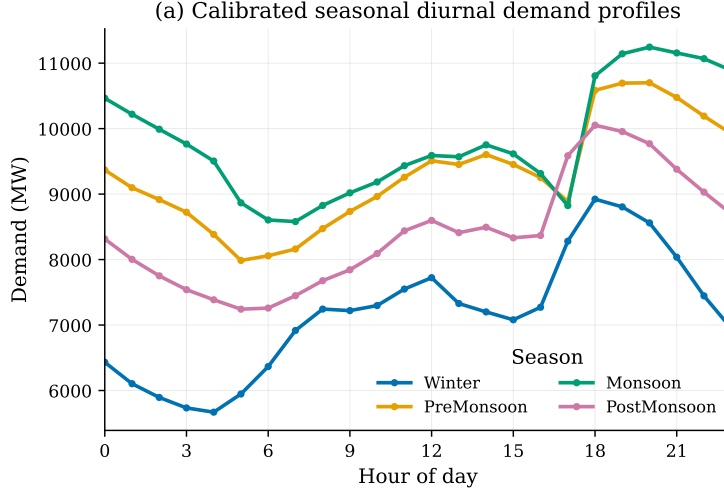


Figure 2: Calibrated seasonal diurnal demand profiles $\bar{D}_s\phi_s(h)$.

7.2 Validation against the historical mix

Figure 4 compares the twin’s seasonal mean dispatch against the recorded mix, with per-source errors in Table 5. The twin tracks the gas-dominated base and the seasonal ordering of coal and hydro. The largest discrepancy is deliberate and economically meaningful: SAED substitutes cheap gas and coal for the expensive liquid-fuel and import generation the historical operation relied on, indicating a recoverable efficiency margin rather than a modelling error.

7.3 Cost, carbon and the growth stress test

Seasonal operating cost and emissions (Figure 5) scale with demand and peak in the monsoon. The demand-growth stress test (Figure 6) propagates the calibrated 7.9% annual growth forward: liquid-fuel peaking energy rises fastest in the monsoon, and the monsoon peak-hour reserve margin is the first to fall below the $\rho = 8\%$ planning threshold (around 2029–2030), identifying the monsoon as the binding season for capacity expansion.

8 Discussion and conclusion

The SAED algorithm reduces seasonal dispatch to a calibrated demand model, a seasonal availability matrix, and a provably optimal merit-order pass (Theorem 1). Driven by ten years of PGCB data, the digital twin reproduces the grid’s seasonal physics and exposes three actionable findings: (i) the historical reliance on liquid-fuel peaking is largely economically avoidable under optimal dispatch; (ii) seasonal generation differences are governed chiefly by hydro and solar availability and by the monsoon demand peak, quantified by the $\alpha_{g,s}$ matrix; and (iii) under the historical growth trend the monsoon reserve margin is the first to breach the planning floor,

Seasonal merit-order dispatch stacks (SAED, year 2025)

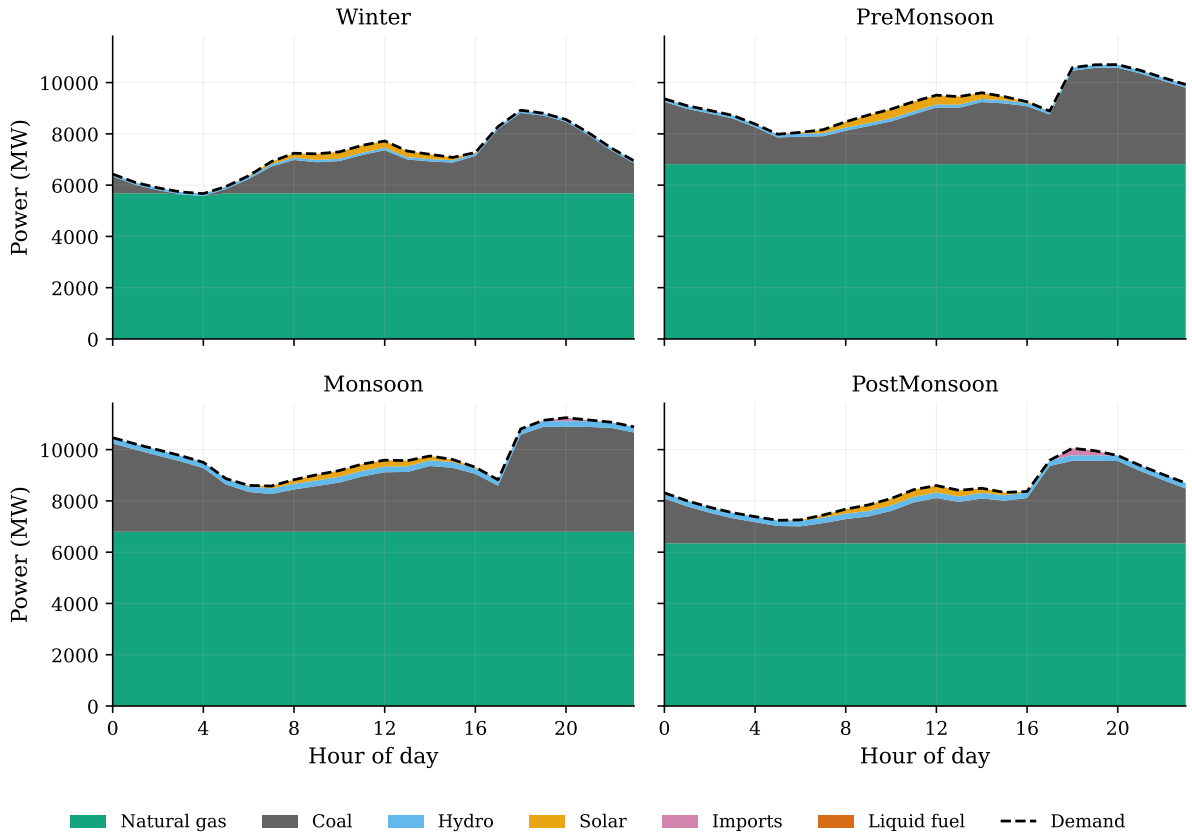


Figure 3: SAED merit-order dispatch stacks for a representative day in each season (base year). The dashed line is demand; the colored bands are the optimal source contributions in merit order.

Digital-twin validation: SAED dispatch vs. historical generation mix

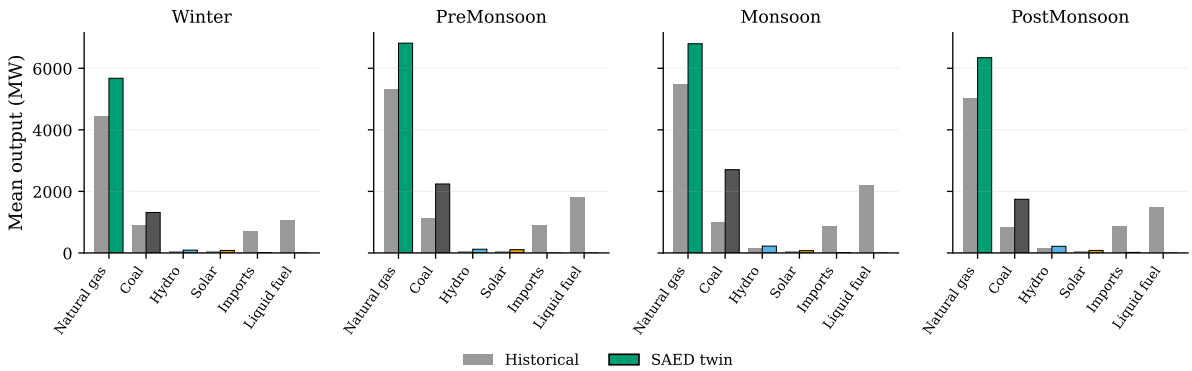


Figure 4: Digital-twin validation: SAED dispatch (colored) versus historical generation (grey) by source and season.

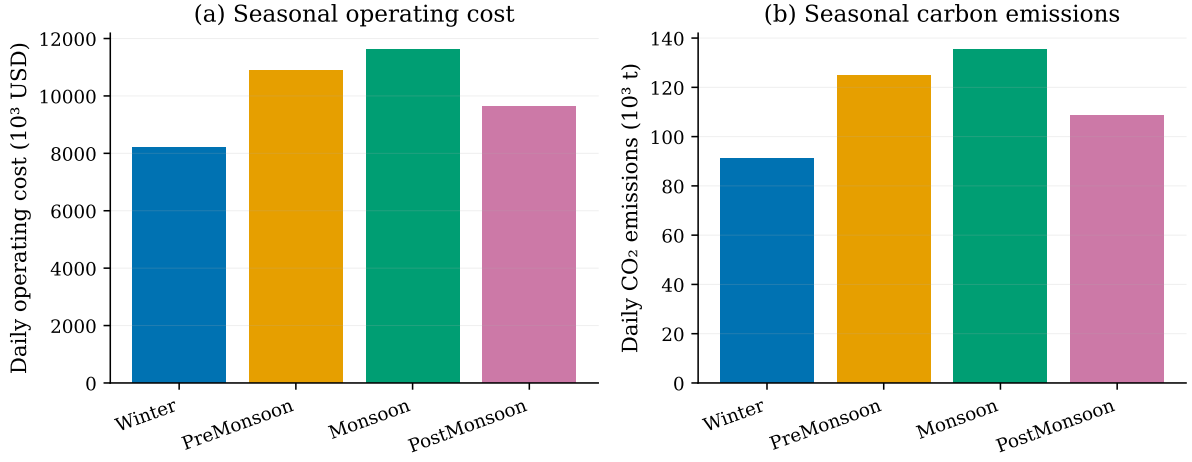


Figure 5: Seasonal daily operating cost (a) and CO₂ emissions (b) from the optimal SAED dispatch.

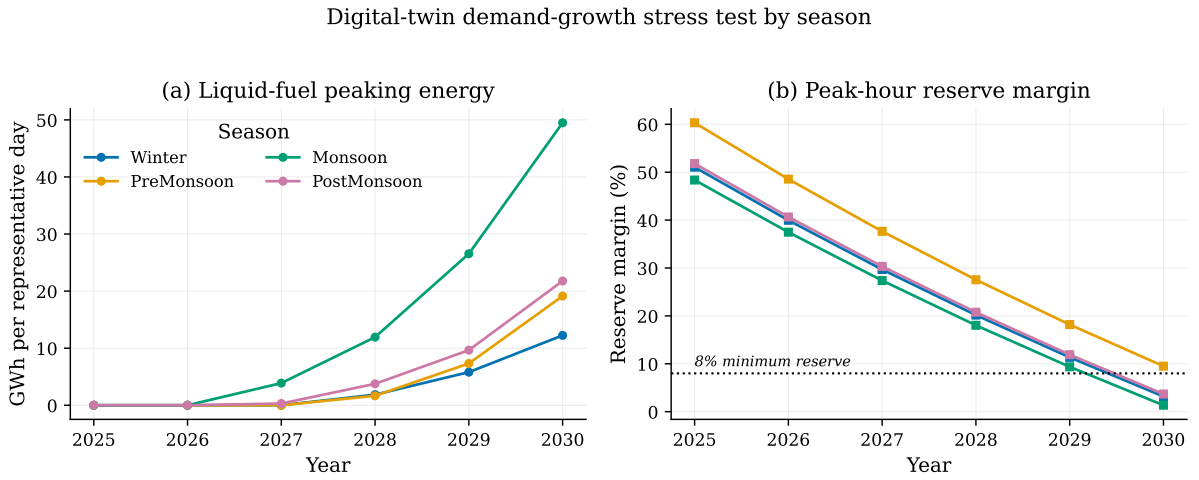


Figure 6: Demand-growth stress test by season: (a) liquid-fuel peaking energy and (b) peak-hour reserve margin. The monsoon crosses the 8% reserve floor first.

making monsoon firm capacity the priority for expansion. The twin is fully reproducible from the accompanying calibration file and source code.